Cyrill Bösch

Laboratory for Intelligent Probabilistic Systems
Princeton University, Princeton, NJ, USA

Email: cb7454@princeton.edu
Web: cyboesch.github.io
ORCID: 0000-0002-1587-7572

Positions

Postdoctoral Researcher, Laboratory for Intelligent Probabilistic Systems, Princeton University, 2024–present

Adviser: Ryan P. Adams

Postdoctoral Researcher, Seismology and Wave Physics, ETH Zurich, 2023–2024

Visiting Researcher, Bertoldi Lab, Harvard University (Host: Katia Bertoldi), 2022

Education

Dr. sc., ETH Zurich, 2019–2024

Thesis: *Topological Matter by Inverse Design.* Adviser: Andreas Fichtner; co-adviser: Marc Serra-Garcia

M.Sc., *Geophysics*, ETH Zurich, 2018

Thesis: *Numerical Homogenization for the Elastic Wave Equation*

B.Sc., *Earth Sciences*, ETH Zurich, 2014

Publications

†Equal contribution.

1. Quantum wave simulation with sources and loss functions. C. Bösch†, M. Schade†, G. Aloisi, S. D. Keating and A. Fichtner. ***Physical Review Research*** 7, 033225 (2025).

2. Adjoint computation of Berry phase gradients. C. Bösch, M. Serra-Garcia, C. Böhm and A. Fichtner. ***Journal of Sound and Vibration*** 619, 119357 (2025).

3. Automated discovery of reprogrammable nonlinear dynamic metamaterials. G. Bordiga, E. Medina, S. Jafarzadeh, C. Bösch, R. P. Adams, V. Tournat and K. Bertoldi. ***Nature Materials*** 23, 1486–1494 (2024).

4. Differences between quantum and classical adiabatic evolution. C. Bösch, A. Fichtner and M. Serra-Garcia. ***Physical Review B*** 110, 064307 (2024).

5. A quantum computing concept for 1-D elastic wave simulation with exponential speedup. M. Schade†, C. Bösch†, V. Hapla and A. Fichtner. ***Geophysical Journal International*** 238(1), 321–333 (2024).

6. Diffuse ultrasound computed tomography. I. E. Ulrich, C. Boehm, A. Zunino, C. Bösch and A. Fichtner. ***The Journal of the Acoustical Society of America*** 151(6), 3654–3668 (2022).

7. Discovery of topological metamaterials by symmetry relaxation and smooth topological indicators. C. Bösch, T. Dubček, F. Schindler, A. Fichtner and M. Serra-Garcia. ***Physical Review***

B 102, 241404 (2020).

Preprints

1. Unifying Physical Backpropagation. C. Bösch, Y. Gediz, C. C. Wanjura and H. E. Türeci. arXiv:2608.11585 (2026).

2. Local Learning Rules for Out-of-Equilibrium Physical Generative Models. C. Bösch, G. Roeder, M. Serra-Garcia and R. P. Adams. arXiv:2506.19136 (2025).

3. Nonlinear Computation with Linear Optics via Source-Position Encoding. N. Richardson, C. Bösch and R. P. Adams. arXiv:2504.20401 (2025).

Patents

U.S. provisional patent application on source-position encoding for nonlinear optical computation, filed through Princeton University; first-named inventor. Pending.

Grants and Fellowships

Princeton University Innovation Grant (USD 200,000, shared between the labs of Ryan P. Adams and Paul Prucnal), 2026

SNSF Postdoc.Mobility Return Grant, EPFL, *Local learning for optical diffusion models* (CHF 115,200), 2025

SNSF Postdoc.Mobility Fellowship, Princeton University, *Probabilistic Mechanical Intelligence* (CHF 124,000), 2023

ETH Doc.Mobility Fellowship, research visit to Harvard University (CHF 2,571), 2022

Awards

ETH Medal for Outstanding Doctoral Thesis, ETH Zurich, 2024

Willi Studer Prize, best M.Sc. graduate in Earth Sciences, ETH Zurich, 2018

ETH Zurich Master's Scholarship Programme, 2015–2017

Invited Seminars

Center for Soft and Living Matter, University of Pennsylvania (groups of Andrea Liu and Douglas Durian), February 2026

Quantum Engineering Group (Hakan Türeci), Princeton University, April 2025

Kollmannsberger Group, Bauhaus University Weimar, April 2024

Selected Talks

2nd Conference on Computing with Physical Systems, Les Houches, France, January 2026

Aspen Summer School on Beyond von Neumann Computing, Aspen Center for Physics, May 2025

Aspen Conference on Computing with Physical Systems, January 2024

Mentoring

Jake Schäfer, undergraduate researcher, Princeton, 2025–present
Yigithan Gediz, doctoral researcher, Princeton, 2025–present
Sowmya Thanvantri, doctoral researcher, Princeton, 2025–present
Malte Schade, doctoral researcher, ETH Zurich, 2024–present; master's researcher, 2023–2024
Arin Mukherjee, master's researcher, Princeton, 2024–present
Nicholas Richardson, doctoral researcher, Princeton, 2024–2025
Yash Sharma, undergraduate researcher, Princeton, 2024–2025